INCH-POUND

MIL-S-22432/31D
28 November 1990
SUPERSEDING
MIL-S-22432/31C
22 August 1974

MILITARY SPECIFICATION SHEET

SERVOMOTOR, TYPE 11SM4C

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Servomotors described herein shall consist of this specification sheet and the issue of the following specification listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-22432.

TABLE I. Requirements.

Requirement	Value	Unit	Tolerance
Electrical:			
Frequency:			
Rated	400	hz	
Test	400	hz	±1%
Range	380 to 420	hz	
Voltage:			
Reference phase, rated	115	volts rms	
Reference phase, test	115	volts rms	±1%
Reference phase, range	103.5 to 126.5	volts rms	
Control phase, rated (series)	36.0	volts rms	
Control phase, test (series)	36.0	volts rms	±1%
Control phase, range (series)	0 to 40	volts rms	
Control phase, rated (parallel)	18.0	volts rms	
Control phase, range (parallel)	0 to 20	volts rms	
Current:			
Reference phase	0.063	amperes rms	maximum
Control phase (series)	0.200	amperes rms	maximum
Control phase (parallel)	0.400	amperes rms	maximum
Power:			
Reference phase	3.7	watts	maximum
Control phase (series)	3.7	watts	maximum
Control phase (parallel)	3.7	watts	maximum
Synchronous Speed	8,000	rpm	
Impedance:			
Reference phase	1010+j1750	ohms	±30%
Control phase (series)	98+j1750	ohms	±30%
Power factor:			
Reference phase	0.50		±1%
Control phase (series)	0.50		±1%
Electrical breakaway torque	0.75	volts rms	maximum
Coupling in control winding	0.75	ratio	minimum

AMSC - N/A

FSC 6105

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TABLE I. Requirements (Continued).

Requirements	Value	Unit	Tolerance
Mechanical:			
No-load speed	6,200	rpm	minimum
Stall torque			
at 23°C	0.65	ounce-inches	minimum
at 125°C	0.55	ounce-inches	minimum
Rotor moment of enertio	0.8	gm-cm²	maximum
Radial play	0.0008	inch	maximum
End play	0.0010	inch	minimum
	0.0035	inch	maximum
Temperature rise	65.0	degrees C	maximum
Time constant	0.02	seconds	maximum
Reversing time	0.03	seconds	maximum
Maximum power output	0.6	watts	minimum
Weight	4.5	ounces	maximum

NOTES:

1. Type 11SM4C servomotors shall be capable of operation in ambient temperatures ranging from -55°C to 125°C.
2. The end cap shall be in accordance with View B, Figure 3 of MIL-S-22432.

TABLE II. Part identifying number variant characteristics.

Part identifying Number 1/	A ±.010	C +.000 -.005	Shaft Style	Termination End	G Maximum
M22432/31-01C	.432	1.062	Splined	Terminal Board	1.629
M22432/31-02C	.432	1.062	Splined	Wire Leads	1.629

1/Part identifying number M22432/31-01C shall be the NATO standard.

1/The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.

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TABLE III. Part identifying number cross-references. 1/

SUPERSEDED PART NUMBERS	NEW PART IDENTIFYING NUMBERS
MIL-S-22432/31C	
M22432/31-1	M22432/31-01C
M22432/31-2	M22432/31-02C

1/All the line items shown in Table III refer to equivalent and interchangeable servomotors of the same type designation modification. Part number changes do not affect form, fit or function of the servomotors listed therein.

Custodians:
Army - AR
Navy - AS
Air Force - 11

Review Activities:
Army - ER, MI, MU
Navy - SH, EC, OS
Air Force - 99
DLA - GS

User Activities:
Army - ME, AV
Navy - MC, CG

Preparing Activity:
Navy - AS

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